

**Project Design Phase-II**  
**Solution Requirements (Functional & Non-functional)**

|               |                            |
|---------------|----------------------------|
| Date          | 2 <sup>nd</sup> July, 2026 |
| Team ID       |                            |
| Project Name  | NUTRITION ASSISTANT        |
| Maximum Marks | 4 Marks                    |

**Solution Requirements**

**Functional Requirements**

| FR No. | Functional Requirement (Epic) | Sub Requirement (Story / Sub-Task)                                                                                                     |
|--------|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| FR-1   | User Registration             | Register using name, email, password, and role (User/Dietitian). Validate user inputs and store encrypted passwords using bcrypt.      |
| FR-2   | User Authentication           | Login with email and password, JWT token generation, secure logout, protected routes, and session management.                          |
| FR-3   | User Profile Management       | View profile, update personal details, change password, manage health information (age, height, weight, dietary preferences).          |
| FR-4   | Client Management             | Dietitian can add, view, update, and delete client profiles, including medical history and nutrition goals.                            |
| FR-5   | Meal Plan Management          | Create, assign, update, delete, and view personalized meal plans containing breakfast, lunch, dinner, snacks, calories, and nutrients. |
| FR-6   | Nutrition Tracking            | Users can log daily food intake, track calories, protein, carbohydrates, fats, fiber, and water intake.                                |
| FR-7   | Progress Monitoring           | Record weight, BMI, calorie intake, and goal completion. Display weekly and monthly progress reports with charts.                      |
| FR-8   | Nutrient Analysis             | Calculate nutrient distribution and provide nutritional summaries and recommendations based on logged meals.                           |
| FR-9   | Admin Dashboard               | Administrator can manage users, approve dietitians, monitor system usage, and generate reports.                                        |
| FR-10  | Reporting & Dashboard         | Display personalized dashboards, nutrition summaries, meal history, and health progress using graphs and charts.                       |

**Non-Functional Requirements**

| NFR No. | Non-Functional Requirement | Description                                                                                                                         |
|---------|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| NFR-1   | Usability                  | The application should provide a simple, intuitive, and responsive interface that is easy to use across desktop and mobile devices. |

|        |                   |                                                                                                                                                                                  |
|--------|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NFR-2  | Security          | Passwords must be encrypted using bcrypt, user authentication should use JWT, and role-based authorization must protect sensitive resources and APIs.                            |
| NFR-3  | Reliability       | The system should ensure accurate data storage, proper error handling, and consistent operation with minimal failures.                                                           |
| NFR-4  | Performance       | API responses should be fast, MongoDB queries optimized, and the application should support smooth navigation with minimal loading time.                                         |
| NFR-5  | Availability      | The application should be available whenever users need it, with reliable database connectivity and minimal downtime.                                                            |
| NFR-6  | Scalability       | The modular MERN architecture should support increasing numbers of users, clients, meal plans, and future enhancements such as AI-based recommendations and mobile applications. |
| NFR-7  | Maintainability   | The codebase should follow a modular structure with reusable React components, organized Express routes, and clear documentation for easy maintenance.                           |
| NFR-8  | Compatibility     | The application should function correctly on modern web browsers such as Google Chrome, Microsoft Edge, Mozilla Firefox, and Safari.                                             |
| NFR-9  | Data Integrity    | All user, meal plan, and progress data should be validated before storage to ensure consistency and accuracy.                                                                    |
| NFR-10 | Backup & Recovery | Regular database backups should be maintained to prevent data loss and enable recovery in case of system failures.                                                               |

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